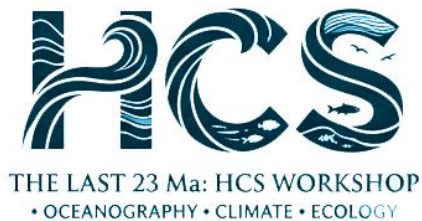


Workshop Abstract & Program Book

The last 23 Ma of the HCS: *oceanographic, climatic and ecological features*

A comprehensive 3-day workshop to discuss the evolution of the Humboldt Current System.



Cologne, Germany
September 2027

Contents

1. [Welcome](#)
2. [Workshop at a Glance](#)
3. [Daily Programme](#)
4. [Scientific Sessions](#)
5. [Keynote Abstracts](#)
6. [Oral Presentation Abstracts](#)
7. [Poster Abstracts](#)
8. [Debate Sessions](#)
9. [Practical Information](#)
10. [Acknowledgements](#)

1. Welcome

The Humboldt Current System (HCS) is one of the most productive marine ecosystems in the world and plays a major role for regional fisheries and food security. Studying its evolution in the past is essential for placing its present-day dynamics into a prospective and longer-term context, particularly as its response to ongoing climate change remains uncertain.

Over the last years, the geological and historical study of the HCS has expanded considerably. Research groups using a wide range of proxies have contributed to our understanding of the different components, mechanisms, and processes associated with the HCS on a range of temporal and spatial scales. These approaches are as diverse as the region itself and include mapping, stratigraphy, sedimentology, geochemistry, ecology, paleontology, climate modelling, and many others.

While this diversity is a strength, it has also resulted in knowledge being developed across relatively disconnected research communities. This is further compounded by the lack of scientific meetings that bring the full range of HCS researchers together. As a result, many people who have been working on related topics for several years, or even decades, have never had the opportunity to meet, beyond sharing literature or exchanging emails.

In spring 2025, we began discussing the need to create a common space for discussion and knowledge exchange. The first concrete steps towards organizing this three-day meeting were taken in fall 2025. The workshop will focus on revisiting the oceanographic, climatic, and ecological evolution of the HCS over the last ~23 million years. Its main goals are to foster scientific exchange, strengthen existing networks, initiate new collaborations, and discuss opportunities for future joint efforts. A further priority is to create a safe discussion space for early career researchers, where they can showcase their work while actively contributing to the broader scientific debate.

As a final message, we would like to emphasize that all participants are expected to engage openly and professionally, listen to different perspectives, and provide others with the opportunity to contribute. Participants are also expected to respect authorship and acknowledge the contributions of others when using or further developing ideas discussed during the workshop. Discriminatory, offensive, or dismissive behavior will not be tolerated. Participation in any workshop activity will be understood as acceptance of and commitment to these principles.

Welcome to the *first paleoHCS workshop*. We hope you will enjoy it!

Patrick and Diana

2. Workshop at a Glance

Wednesday, Sept 23rd		Thursday, Sept 24th		Friday, Sept 25th	
Day 1: Oceanographic Evolution		Day 2: Ecosystem Evolution		Day 3: Proxy-Model Comparisons	
Time	Event	Time	Event	Time	Event
9:00	Welcome & Housekeeping	9:00	Morning Briefing	9:00	Morning Briefing
9:20	Keynote: Frank Lamy	9:05	Keynote: Elisa Malinverno	9:05	Keynote: Pierre Sepulchre
10:20	Morning Talk Block A (2)	10:05	Morning Talk Block A (3)	10:05	Morning Talk Block A (3)
11:00	Coffee Break	11:05	Coffee Break	11:05	Coffee Break
11:20	Morning Talk Block B (3)	11:25	Morning Talk Block B (4)	11:25	Morning Talk Block B (1)
12:20 PM	Group Photo Session			11:45	Debate: Proxy-Model
12:40 PM	Lunch Break	12:45 PM	Lunch Break	12:45 PM	Lunch Break
1:50 PM	Afternoon Reconvening	1:50 PM	Afternoon Reconvening	1:50 PM	Afternoon Reconvening
2:00 PM	Afternoon Talk Block (2)	2:00 PM	Debate: Ecological Dynamics	2:00 PM	Afternoon Talk Block (3)
2:40 PM	Debate: Ocean-Climate Dynamics	3:00 PM	Coffee Break	3:00 PM	Coffee Break
3:40 PM	Coffee Break	3:20 PM	Specific Talks: Landscape of funding opportunities	3:20 PM	Debate: Funding - Actions (dissemination)
4:00 PM	Poster Session: Pitch talks	4:20 PM	Debate: Funding efforts		
4:30 PM	Debate: Record Gaps	5:20 PM	Coffee Break	05:00 PM	General Plenary + Conclusions
5:30 PM	Daily Wrap-up	5:30 PM	Poster Session 2: Open discussion	5:30 PM	Closing Event
from 5:30 PM	BBQ - Social Gathering	7:00 PM	Beer & posters - social gathering (BOB)		

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3. Daily Programme

A complete daily program is also available via the [Workshop website](#).

Day 1: Oceanographic Evolution - Wednesday, Sept 23

09:00 AM - Welcome & Housekeeping

09:20 AM - Keynote: **Frank Lamy**

10:20 AM - Morning Talk Block A (2 Talks)

Reyes-Macaya: *"Major oceanographic changes in the Equatorial and South Pacific during the last 25 ka"*

Toyo: *"Tracing deep circulation and climate variability in the Southeast Pacific"*

11:00 AM - Coffee Break

11:20 AM - Morning Talk Block B (3 Talks)

Hayat: *"Long-term oxygen deficiency and the origin of barren intervals in the core of the Peruvian OMZ inferred from benthic foraminifera"*

Portilho-Ramos: *"Preliminary report on X-ray fluorescence and stable isotope analyses of new deep-sea gravity cores from the Iquique Ridge off Chile and on the Bremen sediment core collection from the SE Pacific"*

Hoogakker: *"Fate of ocean oxygenation in a waRminG wOrld"*

12:20 PM - Group Photo Session

12:40 PM - Lunch Break

01:50 PM - Afternoon Reconvening

02:00 PM - Afternoon Talk Block (2 Talks)

Grunert: *"RV SONNE Cruise SO315: Coupled land-ocean processes as agents of paleoclimatic variability in the Atacama Desert of northern Chile"*

Freire: *"The HCS off the Atacama Desert during the Miocene Climate Transition: Evidence from foraminifera-based proxies"*

02:40 PM - Debate Ocean-Climate Dynamics

03:40 PM - Coffee Break

04:00 PM - Poster Session: Pitch talks (3 mins each)

04:30 PM - Debate Record Gaps

05:30 PM - Daily Wrap-up

From 05:30 PM - BBQ - Social Gathering

Day 2: Ecosystem Evolution - Thursday, Sept 24

09:00 AM - Morning Briefing

09:05 AM - Keynote: **Elisa Malinverno**

10:05 AM - Morning Talk Block A (3 Talks)

Benites-Palomino: *"The last coastal specialists: Functional diversity collapse and ecosystem restructuring in Neogene marine vertebrates"*

Valenzuela-Toro: *"Insights from the Chilean marine mammal fossil record into the evolution of the HCS"*

Chávez-Hoffmeister: *"New Insights into the Bahía Inglesa Formation (Northern Chile)"*

11:05 AM -Coffee Break

11:25 AM - Morning Talk Block B (4 Talks)

Rivadeneira: *"Temperature, time, and transformation: Network-based paleobiogeography of Late Cenozoic bivalves along the Southeastern Pacific coast"*

Medina-Franco: *"Miocene to present turnover of molluscan assemblages: Insights into coastal-marine ecosystem evolution along the Peruvian Margin"*

Nielsen: *"The past of the HCS: Clues from the Neogene onshore record"*

Coenen: *"Miocene diatom biostratigraphy and stepwise Humboldt Current intensification from onshore East Pisco Basin and Offshore ODP records"*

12:45 PM - Lunch Break

01:50 PM - Afternoon Reconvening

02:00 PM - Debate Ecological Dynamics

03:00 PM- Coffee Break

03:20 PM - Specific Talks: Landscape of funding opportunities

04:20 PM - Debate Funding efforts

05:20 PM - Short Break

05:30 PM- Poster Session 2: Open discussion

From 05:30 PM Beer & Posters

Day 3: Proxy–Model Comparisons and Future Perspectives- Friday, Sept 25

09:00 AM - Morning Briefing

09:05 AM - Keynote: **Pierre Sepulchre**

10:05 AM - Morning Talk Block A (3 Talks)

Carré: *"El Niño activity from the Miocene greenhouse to the Anthropocene: Insights from proxy records in the HCS"*

Wennrich: *"Land-ocean coupling as driver of precipitation changes in the central Atacama Desert/ N Chile since the Miocene"*

David: *"Dynamical response of the HCS to past warm climates: a regional modelling approach"*

11:05 AM- Coffee Break

11:25 AM - Morning Talk Block B (1 Talk)

Ochoa: *"A Late Miocene paleobotanical record from the Peruvian Pisco Formation"*

11:45 AM - Debate Proxy-Model

12:45 PM - Lunch Break

01:50 PM - Afternoon Reconvening

02:00 PM - Afternoon Talk Block (3 Talks)

Reyes-Macaya: *"Building a redox-sensitive trace element tool Kit for the Equatorial and Southeastern Pacific: from community science to step-by-step proxy calibration"*

Cardich: *"Benthic foraminifera from a productive bay in central Peru: seasonal community variation to monitor environmental changes in recent decades"*

Susaya-Vicente: *"The HCS off the Atacama Desert during the Miocene Climate Transition: evidence from foraminifera-based proxies"*

03:00 PM - Coffee Break

03:20 PM - Debate Funding - Actions

05:00 PM - General Plenary & Conclusions: Open Mic

05:30 PM - Closing Event

4. Scientific Sessions

Oceanographic Evolution (Day 1)

Session Objective

This session presents key features of the modern Humboldt Current System (HCS) and explores the geological record to assess the HCS evolution from the Miocene to the present. The session is focused on characterizing changes in ocean circulation, upwelling dynamics, and water mass distribution. Implications and effects of the HCS evolution for regional (paleo)climate variability will also be addressed.

The main goal of the session is to establish a shared understanding of the physical evolution of the HCS and identify the major scientific advances and remaining knowledge gaps.

Invited keynote speaker: Frank LAMY (Alfred-Wegener-Institut, Helmholtz-Zentrum für Polar und Meeresforschung, Bremerhaven, Germany)

Ecosystem Evolution (Day 2)

Session Objective

This session examines how long-term oceanographic changes influenced coastal-marine ecosystems, productivity, and biodiversity within the Humboldt Current System and adjacent terrestrial biomes. The session promotes interdisciplinary discussion across paleoecology, paleontology, marine biology, geochemistry, and sedimentology to better understand ecosystem responses to environmental change.

Invited keynote speaker: Elisa MALIVERO (Department of Earth and Environmental Sciences, University of Milano-Bicocca, Italy)

Proxy–Model Comparisons (Day 3)

Session Objective

This session focuses on integrating proxy-based reconstructions with climate and ocean model simulations to improve our understanding of the Humboldt Current System through time. Implications and effects of the HCS evolution for regional (paleo)climate variability will also be addressed. Discussions will identify gaps and opportunities to strengthen collaborations between observational and modelling communities and define priorities for future interdisciplinary research.

Invited keynote speaker: Pierre SEPULCHRE (Laboratoire des Sciences du Climat et de l'Environnement, CNRS-IPSL, France)

5. Keynote Abstracts

Plio-Pleistocene sea-surface temperature changes of the Humboldt Current System and the Pacific Southern Ocean

Frank Lamy ^{*,1}

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¹ Alfred-Wegener-Institut, Helmholtz-Zentrum für Polar und Meeresforschung, Bremerhaven, Germany

In the Pacific sector of the Antarctic Circumpolar Current (ACC) and along the Humboldt Current system (HCS), atmosphere-ocean changes between the high, mid, and low latitudes are strongly linked. Pronounced past changes in surface ocean properties at orbital and longer time-scales indicate an exceptional climate sensitivity of this area.

I am comparing new and previously published Plio-Pleistocene SST and bottom current strength proxy records from the central (IODP Sites U1539 and U1541) and eastern subantarctic ACC off southern Chile (Sites U1542 and U1543) to HCS records from the subtropics (ODP Site 1237 off Peru) and eastern tropical Pacific (Sites 846 (cold tongue) and 1239 (Equatorial Front)). These records are complemented by data from site Geo15016 off northern Chile reaching back to ~1-Ma record.

Taken together, high amplitude glacial-interglacial SST changes in the subantarctic Pacific Southern Ocean are closely coupled to those observed in the subtropical HCS, implying large shifts of the northern ACC system and changing export of subantarctic surface water masses into the HCS. Overall, Pleistocene high-low latitude gradients were enhanced during glacial intervals as the ACC influence extends further into the HCS, likely driven by northward extended subantarctic water masses and stronger intra-Pacific recirculation (Southeast Pacific gyre) as Drake Passage throughflow decreased.

As for glacial-interglacial timescales, high-to-low latitude SST gradients likewise increased on longer time-scales since the Pliocene with the early Pleistocene emergence of the eastern tropical Pacific cold tongue and across the Mid-Pleistocene Transition (MPT). ACC strength at Site U1541 gradually increased between ~1.3 and ~1 Ma coinciding with the early part of the MPT, potentially linked with subsequent SST cooling in the eastern subtropical and tropical Pacific (Sites 1237/1239). This may result from enhanced advection of subantarctic water northward along the HCS as a response to the intensification of the ACC starting during MIS 31.

From phytoplankton to whales: reconstructing the Humboldt Current System through the sedimentary record of the East Pisco Basin

Elisa Malinverno ^{*,1}

Collaborators: Giulia Bosio ^{1,2}, Maria Elena Gastaldello¹, Alberto Collareta², Giovanni Coletti¹, Giovanni Bianucci², Luca Pellegrino³, Giuseppe Marramà³, Giorgio Carnevale³, Mario Urbina⁴ & Claudio Di Celma⁵

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The East Pisco Basin is one of the NW-SE oriented trench-parallel forearc basins that developed during the Cenozoic along the Peruvian convergent margin, due to the subduction of the oceanic Nazca plate beneath the continental South American plate. It hosts an exceptionally rich and well-preserved assemblage of marine vertebrate fossils that make it a world-renowned fossil Lagerstätte. Its sedimentary successions, spanning from the middle Eocene to the Early Pliocene, allows to reconstruct the paleoceanographic history of this area, which is key to understanding of the onset and development of the Humboldt Current.

Here we present the results of our multidisciplinary work in the East Pisco Basin, during the last twelve years. In order to correlate the different outcrops and place all fossils in a chronological context, we built on previous stratigraphic schemes and developed a robust chronostratigraphic framework.

Following a sequence-stratigraphic approach, we identified unconformity surfaces in the field and mapped them over an area of ~1000 km². Such unconformities bound the main stratigraphic sequences named, from older to younger, the Paracas, Otuma, Tunga, Chilcatay and Pisco formations. Minor unconformities allowed identification of units within each sequence, reflecting eustatic cycles. The chronology of each unit was obtained by integrating absolute Ar-Ar age determinations from ash layers, Sr/Sr analyses from carbonate shells and/or shark teeth and integrated biostratigraphy.

Among microfossils, benthic foraminifera, calcareous nannofossils, diatoms, and silicoflagellates provide insights on the paleoceanographic evolution of the area under different climatic contexts. Not least, the abundant exquisitely preserved vertebrate record of the basin enables a complementary perspective on paleoenvironmental evolution, supporting the establishment of a modern-type upwelling regime by the Late Miocene.

Interactions between South American topography and climate since the Miocene: a modeling perspective

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Today, the Andes play a pivotal role in atmospheric and oceanic dynamics by deflecting winds, triggering upwellings, altering large scale atmospheric flow and associated moisture transport, and shaping rainfall patterns across South America. Changes in South American topography during the Cenozoic are therefore expected to have profoundly influenced climate at both regional and global scales. However, quantifying this influence through time has long been challenging because of the limited constraints on paleotopography and lack of suitable models. Over the past decades, substantial progress has been made in reconstructing the timing and magnitude of Andean uplift, opening new opportunities to refine its role in the evolution of South American climate and environments.

This talk will provide an overview of our current understanding of the links between South American topographic evolution and climate during the Neogene. Beginning with the theoretical framework, it will highlight the pivotal role of robust paleogeographic reconstructions for numerical simulations of Neogene paleoclimate over South America. Results from simulations performed with the IPSL-CM5A2 Earth system model for key intervals of the past 20 million years will be presented, with particular emphasis on the South American hydrological cycle and the Humboldt Current System. The talk will also discuss the current limitations of paleoclimate modelling and model–data comparisons for the South American Neogene and will outline future directions to overcome these challenges and improve our understanding of the processes that have shaped the evolution of South American climate and environments.

6. Oral Presentation Abstracts

Talk 1: REYES-MACAYA

Major oceanographic changes in the Equatorial and South Pacific during the last 25 ka: A review

Dharma Reyes-Macaya^{*,1,2,3,4,5}, Babette Hoogakker^{4,5}, Cecilia Ormaza⁵, Camila Saldivia⁶, Sofia Barragán-Montilla^{5,7}, Sebastian Garrido^{4,5}, Laetitia Pichelvin⁸, Sheetal Samal⁶, Raul Tapia⁹, Alexis Castillo Bruna¹⁰, Katya Canal⁴, Renato Salvattecí¹¹, Iván Hernández-Almeida¹², Thomas A. Ronge¹³, Cyrus Karas¹⁴, Consuelo Martí Nez Fontaine¹⁵, Praxedes Muñoz¹⁶, Valentina Flores-Aqueveque¹⁷, Maria Toyos³, Martina Hollstein³, Lester Lembke-Jene¹⁸, Rodrigo Da Costa Portilho-Ramos³, Elisabeth Teca-Oliva¹⁹, William Gray¹⁹, Elisabeth Michel¹⁹, Nicolas Glock²⁰, Jorge Cardich²¹, Helen Bostock²², Claudia Aracena²³, Edgardo Flores²⁴, Giuseppe Siani¹⁸, Frank Lamy¹⁸, Krystel Susaya Vicente²⁵, Margarita Marchant²⁶, Hidetaka Kobayashi²⁷, Allison Jacobel²⁸, Jean Lynch-Stieglitz²⁹, Julia Gottschalk⁷, Naoufel Haddam¹⁹, Manuel Torres-Godoy², Alexander Galan^{10,25}, Karen Sanzana Chavez², Eugenia M. Gayo^{31,32}, Laura Farias^{2,30,31}, Cristian Chandia², Marco Salamanca², Linda Barranco²⁵, Mahyar Mohtadi³, Carina Lange², Catherine Davis³³, Samantha Bova³⁴, Andreas Lü Ckge³⁵, Patrick Grunert⁶, Andrea Jaeschke^{6,36}, Olga Koukousioura³⁷, Julio Sepulveda²⁴, Shankle Madinson³⁸, Jorge Valdés³⁹, Antonio Maldonado²⁵, Paola Cardenas^{2,40}, Sergio Contreras⁴⁰, Dierk Hebbeln¹

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The Humboldt Current System in the Equatorial and Southeast Pacific (ESEP) has dynamically regulated Earth's climate, yet its historical response to atmospheric forcing remains complex. This study presents a high-resolution multiproxy synthesis, integrating records of sea surface temperature, export productivity, and water-column properties, spanning the last 25,000 years to evaluate the mechanisms driving regional circulation, deep water ventilation, and exported productivity. During the Last Glacial Maximum, northward-shifted Southern Westerly Winds and expanded sea-ice highly stratified the deep ocean, turning the ESEP into a massive trap for old, unventilated carbon. Deglacial ventilation was not uniform; the ESEP acted as a "Shadow Zone" retaining its isolated carbon reservoir significantly longer than the western Pacific. During the Younger Dryas, the Antarctic Intermediate Water finally flushed the Southeast Pacific, acting as a rapid conduit to drive CO₂ outgassing. Furthermore, the Holocene experienced a critical regime shift from stable, stratified Early Holocene conditions to the highly variable, ENSO-dominated coastal upwelling system observed today, driven by an intensified Walker Circulation. Ultimately, this review establishes that the ESEP acted as a dynamic

capacitor for the global carbon cycle, providing a vital baseline for forecasting how the Humboldt Current System will respond to future anthropogenic warming.

Talk 2: TOYOS

Tracing deep circulation and climate variability in the Southeast Pacific (DEEP-PAC)

María H. Toyos^{*,1}

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Pacific Deep Water (PDW), the largest deep-water mass in the global ocean, flows southward along the Peru–Chile margin and enters the Southern Ocean via Drake Passage. Despite its critical role in global circulation and climate, no reconstructions of PDW flow speeds exist for the southeastern Pacific across orbital and suborbital timescales. This limits our understanding of how deep circulation responded to past climate extremes, such as the Last Glacial Maximum, the last interglacial (MIS 11), and events like the Antarctic Cold Reversal, especially under varying topographic influences.

This project will reconstruct bottom-current variability over the past 500,000 years using sediment cores along the Peru–Chile margin, with multi-resolution sampling to capture both long-term trends and abrupt changes. Key proxies include CT scanning for depositional structures, XRF-derived Zr/Rb ratios, and Sortable Silt to infer current strength. Results will be compared with present-day circulation from the GLORYS12 reanalysis to contextualize past dynamics within modern oceanographic frameworks.

This work will deliver the first long-term reconstruction of PDW flow in the southeastern Pacific, clarifying how deep circulation evolved in response to climate change and advancing our understanding of ocean–atmosphere interactions over glacial–interglacial cycles.

Talk 3: HAYAT

Long-term oxygen deficiency and the origin of barren intervals in the core of the Peruvian oxygen minimum zone inferred from benthic foraminifera

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The Peruvian oxygen minimum zone (OMZ) is among the most intense low-oxygen systems in the ocean, yet quantitative reconstructions of its past oxygenation remain scarce, especially from its core. Here, we reconstruct bottom-water oxygen (BWO) over the last 25 kyr using the extended Benthic Foraminiferal Assemblage index (BFA_{ex}) applied to cores G10 and G14 (312 and 390 m water depths, 14°S) and extend it to 650 kyr BP using published benthic foraminifera counts from ODP Site 680B (252 m, 11°S). Out of 166 samples analyzed, 88 were barren of foraminifera. Microtomographic imaging reveals foraminifera shaped dissolution textures, agglutinated tests, and gypsum crystals, showing that foraminifera were deposited but later dissolved during transient ventilation events rather than ecologically excluded. The BWO reconstruction indicates persistently near-anoxic waters, interrupted by brief oxygenation pulses during Heinrich Stadial 1 probably linked to strong El Niño activity. The 650-kyr record shows no glacial-interglacial cyclicity, with 58% of samples showing below 0.1 mL L⁻¹ BWO, establishing extreme deoxygenation as a structural feature of the OMZ core. Comparison with redox-sensitive metals and $\delta^{15}\text{N}$ reveals decoupling during glacial intervals, likely reflecting Antarctic Intermediate Water influence. BWO conditions in the Peruvian OMZ therefore are impacted by local productivity and remote ventilation.

Talk 4: PORTILHO-RAMOS

Preliminary report on X-ray fluorescence and stable isotope analyses of new deep-sea gravity cores from the Iquique Ridge off Chile and on the Bremen sediment core collection from the SE Pacific

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During the Carnival Cruise along the northern Chilean margin, 12 gravity cores were recovered from the Iquique Ridge at water depths ranging from approximately 3,000 to 4,500 m and are currently stored at MARUM. X-ray fluorescence (XRF) analyses were conducted to establish a preliminary stratigraphic framework and support the development of age models for the sediment cores from this area. Preliminary results reveal contrasting downcore Fe/Ca patterns among the studied records. Cores GeoB27120, GeoB27123, GeoB27124, and GeoB27125 display similar Fe/Ca variations, with values generally ranging between 4 and 12. In contrast, cores GeoB27117 and GeoB27118, recovered at similar water depths and locations, exhibit substantially lower Fe/Ca values, typically below 1. Foraminiferal investigations performed for stable isotope analyses indicate that much of the observed Fe/Ca variability may be linked to carbonate dissolution. Well-preserved foraminifera are generally only present in intervals where Fe/Ca values decrease below approximately 5, preventing the establishment of continuous isotope-based chronologies. Particularly intriguing is the observation that –although all cores were recovered from above the carbonate compensation depth– some cores appear to show stronger dissolution and others do not, despite having been taken from similar water depths. Ongoing stable isotope analyses of cores GeoB27117 and GeoB27118 aim to investigate further the mechanisms controlling carbonate preservation and sedimentary variability in the study area. Furthermore, a brief overview about sediment cores from the SE Pacific being available in Bremen will be provided.

Talk 5: HOOGAKKER

Fate of ocean oxygenation in a waRminG wOrld

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The Fate of ocean oxygenation in a warming wOrld (or 'FARGO') project has the aim to improve our understanding of Pacific Ocean oxygenation, investigating the impact of the closure of the Central American Seaway (CAS), and the effects of warmer climates. For this project we have carried out further development of various proxy methods, using sediment samples from across the SE Pacific margin. Among other locations, we also utilize sediment from ODP Sites 1236 and 1237, which are under the influence of the Humboldt current.

Interpretation of work relating to the mid-Pliocene Warm Period is currently underway, as are reconstructions for key interglacials 9 (warmer) and 13 (cooler). Our proxy reconstructions and modelling efforts confirm that the closure of the CAS led to widespread intensification of deoxygenation of subsurface and deep waters in the Pacific Ocean. Deeper waters likely became less oxygenated ~ 11 Ma ago, while intensification of the OMZs is estimated to have occurred around 7-8 Ma ago, the stages timings related to CAS sill depths.

Talk 6: GRUNERT

RV SONNE Cruise SO315: Coupled land-ocean processes as agents of paleoclimatic variability in the Atacama Desert of northern Chile (SE Pacific)

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RV *SONNE* Cruise SO315 (Oct. 8 – Nov. 17, 2025) recovered new water column samples and sediment cores together with physicochemical and hydroacoustic data along the northern Chilean continental margin and offshore seamounts (18°30'–24°30'S). To establish a modern-day reference framework, vertical CTD profiles of physicochemical parameters of the water column were accompanied by a comprehensive water and plankton sampling program. To identify pristine sedimentary sequences on the tectonically active continental margin and seamount areas, an extensive hydroacoustic survey of the seafloor was conducted. To recover the desired paleoceanographic and paleoclimatic archives, surface and subsurface sediments were sampled along depth transects from all work areas. Highlights include the exceptional recovery of a ~ 17 m long core of Holocene sediments in Mejillones Bay and transects across the submarine fan of the Río Loa. The recovered material represents the most comprehensive sedimentary archive off northern Chile.

The samples and data will 1) enable baseline studies on the regional oceanography, biogeochemical cycles and proxy calibration; and 2) establish new paleoceanographic and paleoclimatic records from late Quaternary marine archives. They will serve as the foundation for the study of coupled processes between the subtropical south-eastern Pacific and southern Atlantic and the spatial and temporal paleoclimatic dynamics of the Atacama Desert on orbital and suborbital timescales.

Talk 7: FREIRE

The Humboldt Current System off the Atacama Desert during the Miocene Climate Transition: Evidence from foraminifera-based proxies

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Palaeoceanographic archives of the Humboldt Current System (HCS) offshore the Atacama Desert are fundamental for understanding the evolution of Atacama's hyper-aridity. Mid-Miocene changes in HCS properties were investigated from the sediment outcrop Caleta Herradura, located on the Mejillones peninsula of northern Chile. The chronology is based on magnetostratigraphy, tephrochronology, oxygen isotope stratigraphy of benthic foraminifera, and planktonic foraminifera biostratigraphy. The age-model suggests that the investigated section spans from 14.4 to 12.8 Ma, including the Miocene Climate Transition (MCT; 14.7–13.8 Ma). Individual Foraminiferal Analysis (IFA) – stable oxygen and carbon isotopes – of various foraminiferal species was performed to reconstruct relative depth habitats. To estimate changes in ocean temperature and water-column stratification, Mg/Ca ratios of planktonic foraminifera from distinct depth habitats (*Trilobatus immaturus*, *Dentoglobigerina altispira*, *Dentoglobigerina venezuelana*) were determined by LA-ICP-MS. Productivity was estimated using stable carbon isotopes of benthic foraminifera and relative abundances of the planktonic foraminifer *Globigerinita glutinata*. IFA illustrates that Mg/Ca temperatures are derived from the upper and lower mixed layer and sub-thermocline depths. The results highlight surface cooling during the study interval, reflecting either global MCT changes or the onset of coastal upwelling, as it is decoupled from the sub-surface and correlated to increased water mixing and productivity.

Talk 8: BENITES-PALOMINO

The last coastal specialists: Functional diversity collapse and ecosystem restructuring in Neogene marine vertebrates

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Late Miocene marine vertebrate communities occupied a broader range of ecological niches and functional trait combinations than those observed in modern oceans. Coastal ecosystems supported numerous specialized taxa, including the tusked dolphin *Odobenocetops*, the aberrant kogiid *Scaphokogia*, the aquatic sloth *Thalassocnus*, and diverse odontocetes exhibiting feeding adaptations and cranial morphologies without modern analogues. Many of these taxa disappeared during the Miocene–Pliocene transition, a period marked by global cooling, sea-level fluctuations, and major reorganization of coastal marine environments. While taxonomic turnover during this interval is well documented, its consequences for functional diversity and ecosystem structure remain poorly understood. This project investigates patterns of functional diversity loss and ecological restructuring across the Miocene–Pliocene transition using marine vertebrate assemblages from Peru, Chile, California, Argentina, and the North Sea Basin. Focusing on odontocete cetaceans as a model group, we combine three-dimensional digital morphology, geometric morphometrics, and functional trait analyses to quantify changes in cranial function and feeding ecology through time. We test the hypothesis that environmental change selectively eliminated coastal specialists, resulting in a contraction of marine vertebrate ecospace and the loss of ecological roles absent from modern ecosystems. The results will provide new insights into the assembly of modern marine vertebrate communities globally.

Talk 9: VALENZUELA-TORO

Insights from the Chilean marine mammal fossil record into the evolution of the Humboldt Current System

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Fossil marine mammals from Chile provides insights into the evolution of the Humboldt Current System and its associated marine ecosystems over the last ~7 million years. Marine mammal remains are predominantly recovered from Late Miocene deposits of the Bahía Inglesa Formation in the Atacama Desert. The fauna comprises morphologically and ecologically diverse taxa, including rorqual whales, non-delphinid odontocetes, phocids, sirenians, and aquatic sloths, pointing to complex trophic networks and ecological guilds with no modern analogues in the region. The Pliocene record is geographically widespread, with remains from the Bahía Inglesa, Coquimbo, and Horcón formations in northern and central Chile and an isolated record in Patagonia. Although more fragmentary, this record is represented by aquatic sloths, cetaceans, and phocids, and documents the co-occurrence of extinct Miocene lineages with more derived fauna (e.g., early delphinids). Moreover, this interval was marked by an extensive marine mammal faunal turnover in the South Pacific, likely resulting from global transformations in marine ecosystems. Fossils from the Pleistocene are rare and only referred to taxa closely related to extant species. These records suggest that modern ecological configurations were already established during the Pleistocene and document relatively recent biogeographic transformations of the marine mammal communities.

Talk 11: RIVADENEIRA

Temperature, time, and transformation: Network-based paleobiogeography of Late Cenozoic bivalves along the Southeastern Pacific coast

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Sea surface temperature (SST) is a primary driver of marine biogeographic patterns in modern oceans, yet its long-term influence on paleobiogeographic dynamics remains insufficiently explored. This study focuses on the southeastern Pacific (SEP) during the late Cenozoic, an interval characterized by pronounced climatic variability and environmental transitions. We applied network theory to both fossil and modern bivalve distributional data across approximately 23 Myr. By constructing bipartite biogeographic networks, we uncovered remarkably robust spatial and temporal patterns that persist even after accounting for sampling biases. Notably, the analyses reveal a trend of decreasing modularity and a reduction in the number of distinct bioregions over the last 3 million years. At the species level, thermal optimum and thermal range emerged as the most significant predictors of biogeographic position. Topological network metrics further highlighted that fossil sites functioned as either peripheral nodes or connectivity hubs, whereas the majority of species occupied peripheral roles modulated by their niche breadth. Overall, these findings demonstrate that SST and associated paleoenvironmental shifts have been dominant forces shaping SEP bivalve biogeography. The network-based approach provides novel insights into thermal forcing mechanisms and evolutionary community dynamics, with important implications for forecasting marine biodiversity responses to ongoing climate change.

Talk 12: MEDINA-FRANCO

Miocene to present turnover of molluscan assemblages: Insights into coastal-marine ecosystem evolution along the Peruvian Margin

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Over the past 10 million years, coastal-marine settings along the Peruvian Margin have undergone profound geographic and oceanographic transformations, driving extensive changes in coastal-marine communities. While mollusk taxonomy research is slowly being integrated into ecosystem-wide analyses, a long-term chronostratigraphically controlled analysis of molluscan diversity and compositional changes has not been undertaken for this region. We compiled a database covering 152 species, 97 genera, and 51 families of mollusk fossils from the Peruvian Margin (13–16°S) to assess long-term diversification patterns and faunal turnover from the late Miocene to the present. We identified two distinctive molluscan assemblages. The first, dating to the late Miocene (10–6 Ma), underwent a substantial shift during the Mio-Pliocene transition (6–4 Ma), culminating in a second assemblage more akin to modern counterparts and exhibiting greater genus richness. This turnover was driven by peaks in bivalve origination (6–5 Ma) and elevated extinction rates for gastropods (6–5 Ma) and bivalves (5–4 Ma). Ecological analyses revealed no single trait consistently changed during this interval, indicating a broad reorganization of ecological strategies. We propose this turnover reflects geomorphological changes related to Andean uplift, the disappearance of semi-embayments, and sea-level rise.

Talk 13: NIELSEN

The past of the Humboldt Current System: Clues from the Neogene onshore record

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Climate change is one of the major drivers of biodiversity and one of the easier-to-reconstruct parameters of past environments. Although progress has been made in recent years, little is known about the onset and evolution of the Humboldt Current System.

Lack of close-by deep ocean drilling sites makes onshore outcrops along the Pacific margin of South America the only source to understand timing and nature of near-coast oceanographic changes and the resulting faunal responses. We are currently in the process of describing and documenting faunas of localities with aragonite preservation, to be able to reconstruct the evolution of mollusks through time, combined with reconstruction of past temperatures and bathymetry. Mollusks are by far the most abundant and diverse group of fossils, being also sensitive to environmental parameters. Several new records improve constraints on stratigraphic and geographic ranges and include taxa without previous fossil record. Analysis of richness and relative abundance as well as faunal compositions help to reconstruct the paleoenvironment and define biogeographic structure.

Talk 14: COENEN

Miocene diatom biostratigraphy and stepwise Humboldt Current intensification from onshore East Pisco Basin and Offshore ODP records

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Diatom records from the East Pisco Basin (EPB) of Peru and Ocean Drilling Program (ODP) sites provide a high-resolution archive of the evolution of the Humboldt Current System (HCS) over the last ~23 Myr. Here we synthesize diatom biochronology and productivity proxies from ODP Holes 682A and 688E to refine Miocene diatom zonations for the EPB, establish a nearly continuous offshore reference section, and strengthen correlations to fragmentary onshore records. The record captures a complete succession of low-latitude and northeastern Pacific Miocene diatom zones, punctuated by two key gaps: a dissolution/hiatus interval during the Middle Miocene Climatic Optimum (~16.5 – 14 Ma) and a likely earliest Miocene hiatus (~23.4 – 21.8 Ma). Offshore-onshore comparisons reveal stepwise intensification of HCS primary productivity through the middle-late Miocene, including the onset of enhanced diatom deposition (~14.2 – 13.8 Ma), productivity reorganization near ~12.9 Ma coincident with a major fall in sea level, a major increase in diatom sedimentation rates beginning at ~10.4 – 10.0 Ma, and further enhanced diatom deposition during transgressive Pisco-2 (~8.4 Ma) and the Late Miocene Biogenic Bloom (8.0 – 4.5 Ma). We discuss how tectonics, eustasy, and ENSO-like variability (7–6 Ma) shaped HCS oceanography and ecosystems, and outline opportunities for community synthesis.

Talk 15: CARRÉ

El Niño activity from the Miocene greenhouse to the Anthropocene: Insights from proxy records in the Humboldt Current System

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El Niño–Southern Oscillation (ENSO), the primary driver of interannual climate variability worldwide, produces its most dramatic perturbations in the Humboldt Current System (HCS). Changes in ocean–atmosphere dynamics during extreme El Niño events generate large sea-surface temperature anomalies (up to 8°C), heavy rainfall and flooding, and suppress the upwelling of deoxygenated, nutrient-rich subsurface waters, with major impacts on marine and coastal ecosystems. ENSO is therefore a key geophysical driver influencing coastal sedimentology, geomorphology, ecosystem structure, and evolutionary trajectories. Reconstructing past changes in its activity is critical for understanding not only its response to variations in Earth’s radiative forcing but also the long-term evolution of ecosystems.

Here we present an overview of the archives and techniques used to reconstruct past changes in ENSO activity in the HCS, and of the most recent reconstructions across a broad range of climate conditions, including the Late Miocene greenhouse, the last interglacial warm period, the last deglaciation, the Holocene, and the last millennium. We show how paleo-ENSO data from the HCS constrain future climate projections.

Talk 16: WENNRICH

Land-ocean coupling as driver of precipitation changes in the central Atacama Desert/ N Chile since the Miocene

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The Atacama Desert is one of the oldest and driest deserts on Earth. Its long-term hyperaridity is attributed to the Andean rain shadow and the evolution of Humboldt Current System (HCS) during the Cenozoic. However, due to the scarcity of continuous climate archives, the Cenozoic precipitation history of the Atacama is only poorly investigated. Furthermore, the timing, forcings, and feedbacks of land-ocean coupling between the Humboldt Current System (HCS) and the Atacama hydroclimate remain poorly understood.

As part of the CRC1211 (Collaborative Research Center 1211): “Earth evolution at the dry limit”, our project focuses on long-term precipitation changes in the hyperarid core of the Atacama Desert and its oceanic drivers. The investigated continental records in the Coastal Cordillera indicate significant precipitation changes since the Mid Miocene, with longer lake phases and significant vegetation cover during pluvial phases.

The findings were linked with new results from well-dated marine outcrops along the Atacama coast that cover similar timeframes. Geochemical proxy records of foraminiferal tests from these outcrops enable the reconstruction of properties of the HCS, in particular sea surface temperatures and coastal upwellings. The integration of data from the synchronized marine and terrestrial records enables to directly infer patterns of precipitation response to HCS changes.

Talk 17: DAVID

Dynamical response of the Humboldt Current System to past warm climates: A regional modelling approach

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The Humboldt Current System (HCS) generates the world's highest-productivity and coolest tropical surface waters due to coastal upwelling, which brings cold, nutrient-rich water to the surface. However, global climate models (GCM) exhibit significant biases in this region. These biases are believed to contribute to the discrepancy between model projections and observations in the tropical Pacific. Paleoclimates can be used as real-world experiments to understand how the HCS responds to a broad range of radiative forcing. We propose investigating the HCS's response to past warm climates by performing dynamical downscaling of paleoclimate simulations from the IPSL-CM6A-LR model using the CROCO regional model. This approach is expected to mitigate GCM biases and enhance the mesoscale dynamics representation. We will examine three warm paleo periods: the mid-Holocene (6 ka), the Eemian(127 ka), and the mid-Pliocene period (3.5 Ma). The first two periods will inform us about the HCS's sensitivity to insolation forcing. The mid-Pliocene period can be used as an analog for a future scenario with a CO₂ concentration of 400ppm. By comparing these results with paleoclimate proxy reconstructions, we expect to improve our understanding of the eastern tropical Pacific's sensitivity to radiative forcings and strengthen our confidence in future HCS projections.

Talk 18: OCHOA

A Late Miocene paleobotanical record from the Peruvian Pisco Formation

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Xeric ecosystems are particularly vulnerable to global warming. Changes in water availability and precipitation patterns may contribute to the development of new floras and biomes, transforming landscapes and existing biotic interactions. The Late Miocene, characterized by warm climates and pCO₂ levels comparable to those of today (~400 ppm), provides an opportunity to understand potential ecosystem responses under steady-state warming scenarios. In this study, we present the first known record of fossil pollen, leaves, and wood from the Late Miocene (8-6 million years ago) of the Peruvian coast (15°S), where today the landscape is dominated by a desert with little to no vegetation.

The paleobotanical record reveals a diverse community dominated by elements typical of modern seasonally dry forests rather than desert vegetation, including a mixture of taxa from lowland forest and Andean biomes. Recalibrated precipitation estimates indicate coastal rainfall values four times higher than those of the present day. Meanwhile, the western slopes experienced significantly wetter conditions during the austral summer compared with the present. These findings suggest a substantial ecosystem shift that enabled the development of a greener, forested landscape along the central coastal region of Peru in response to the warm climate of the Late Miocene, while also influencing the development of the coastal-marine communities of that time.

Talk 19: REYES-MACAYA

Building a redox-sensitive trace element tool Kit for the Equatorial and Southeastern Pacific: From community science to step-by-step proxy calibration

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Reconstructing past marine deoxygenation within the highly productive Humboldt Current System requires the robust calibration of redox-sensitive trace element proxies (e.g., U, V, Mo, Re, Cd) in marine sediments. However, accurately interpreting these proxies along the South American margin relies on isolating authigenic enrichment from the complex terrestrial lithogenic background delivered by Chilean river networks and the Atacama Desert. To address this, we establish a high-resolution biogeochemical baseline mapping trace metals and stable isotopes across a vast latitudinal gradient (12°-50°S). Crucially, this terrestrial end-member is co-created through horizontal community science methodologies, for local communities to autonomously sample and monitor their watersheds. Building upon this framework, future research can apply these calibrated baselines to refine trace metal proxies in foraminifera and accurately quantify historical marine deoxygenation in the Humboldt Current. Furthermore, this pre-industrial lithogenic knowledge serves to trace the modern impact of invasive species in coastal regions, which act as “biogeochemical pumps” transporting marine-derived nutrients and metals upstream and accumulating these tracers along their migration routes. By bridging lithogenic and marine geochemistry, it not only refines past oceanic oxygen proxies but translates empirical data into policies. Ultimately, this approach addresses critical regulatory gaps in Chilean sediment monitoring and fosters community-led sustainable management from the Andes to the coastal ocean.

Talk 20: CARDICH

Benthic foraminifera from a productive bay in central Peru: Seasonal community variation to monitor environmental changes in recent decades

Jorge Cardich^{*,1}, Lander Merma-Mora¹, Dennis Romero-Chuquival², Nohelia Cirilo-Hurtado¹, Ivana Guevara-Kihien¹, Gustavo López-Meza¹, Bruno Turcq³, Matthieu Carré³, Alexander Pérez¹, Dimitri Gutiérrez²

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Shallow subtidal environments of Peru are subjected to the influence of both coastal upwelling and anthropogenic pressure (industrial fisheries, oil companies, urban discharges, etc.). As a result, the bays of Peru are highly productive and oxygen deficient. Benthic foraminiferal communities are mainly composed of a low diversity of small calcareous species. Samanco Bay is one of the most productive bays in Peru. As human perturbations are still not considerable here (only aquaculture since late 1990s), oceanographical effects on the coast might be traceable through foraminifera. We aimed to characterize seasonal differences in bottom redox conditions by studying the community structure of benthic foraminifera. We then applied this information to reconstruct changes during the last half century (SAM core, 14 m depth).

The bottom waters were hypoxic during all the study, but water column thermal stratification was higher in Dec-24. pH values ranged from 7.2 to 7.6. The porewater sulfides and nutrients concentrations were higher in Apr-2025. Small species, such as *Bolivina costata* and *Buliminella elegantissima*, dominated the inner bay (<15 m). The outer bay (>20 m) was characterized by high abundances of the “large” *Nonionella auris*, which entered the inner bay in winter. Accompanying species, such as *Discorbis peruvianus*, *Virgulinella fragilis* and *Buccella peruviana* were more common in the inner, less dynamic station. Sediments from the core were banded and did not present perturbations. Older samples were characterized by equitative assemblages, with *B. peruviana*, *B. costata* and *N. auris*. From centimeter 30 to the core-top, *B. costata* dominated the assemblage.

Benthic foraminifera assemblages in Samanco Bay are low diverse and quite constant over time. Dominant species are adapted to environmental variability, and the assemblages serve to reflect the changes in the bay at seasonal and decadal time scales.

Talk 21: SUSAYA-VICENTE

Marine biogeographic provinces of the Southeastern Pacific: Evidence from living benthic foraminifera

Krystel Susaya-Vicente^{*1}, Reinaldo Rivera², Alexis Castillo-Bruna³, Lander Merma-Mora⁴, Jorge Cardich⁴, Joachim Schönfeld⁵, Nicolaas Glock⁶, Sebastian Garrido⁷, Babette Hoogakker⁷, Margarita Marchant⁸

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Marine biogeographic provinces along the Southeastern Pacific margin have historically been defined based on macroinvertebrate distributions, recognizing major latitudinal transitions associated with tropical, warm-temperate, and cold-temperate/subantarctic domains. However, benthic foraminifera, despite representing a dominant component of marine sedimentary ecosystems, have been scarcely incorporated into regional marine biogeographic frameworks.

Unlike many macrofaunal groups, benthic foraminifera exhibit strong ecological sensitivity to environmental thresholds such as bottom-water oxygenation, organic matter flux, water depth, and sediment characteristics. Furthermore, most previous studies have relied on total (living and dead) assemblages, potentially obscuring ecological boundaries through sedimentary reworking and taphonomic processes. In contrast, living assemblages provide a direct ecological snapshot of communities responding to contemporary environmental conditions.

Using a species distribution database and a biogeographic regionalization approach, we evaluated the spatial organization of living benthic foraminiferal assemblages along the western margin of South America. Our results identified ten distinct biogeographic regions across the Southeastern Pacific, revealing not only clear latitudinal discontinuities but also a pronounced bathymetric and cross-shelf structuring component.

7. Poster Abstracts

- [Poster 1: Giulia BOSIO](#)
Fossil Evidence for the Strengthening of the Humboldt Current During the Late Miocene (East Pisco Basin, Peru)
- [Poster 2: Sebastián GARRIDO](#)
The Warm, the Cool, and the Deep: Contrasting deep-water oxygenation during MIS 9 and MIS 13 in the Southeast Pacific
- [Poster 3: Paleo3 Journal](#) - Special Issue on SA
From Land to Ocean: South American climate variability from the Miocene to Holocene
- [Poster 4: Javiera Andrea ROCO-VILLABLANCA](#)
Scaphopoda of the Miocene of Chile
- [Poster 5: Vicente GUERRERO](#)
Paleobiogeographic patterns of Lower Miocene (17-18 Ma) marine faunas from south-central Chile
- [Poster 6: Krystel SUSAYA-VICENTE](#)
Environmental Gradients Shape Functional Diversity in Living Benthic Foraminifera
- [Poster 7: Juan-Felipe MONTENEGRO](#)
Middle Miocene Woods from the Pisco Formation
- [Poster 8: Tom DeVRIES](#)
The "Oligocene Gap" in the East Pisco Basin (Peru): An Opportunity to Predict the Paleogene-to-Neogene Evolution of the Humboldt Current System
- [Poster 9: Aparna PRASANNAKUMAR](#)
The Evolution of Humboldt current system in AWI-ESM
- [Poster 10: Camila Fernanda SALDIVIA](#)
Bathymetric changes in the Seno Hoppner Formation, Southern Chile, across the Pliocene–Pleistocene boundary based on benthic foraminifera

Poster 1: BOSIO

Fossil Evidence for the Strengthening of the Humboldt Current During the Late Miocene (East Pisco Basin, Peru)

Giulia Bosio^{*,1}, Eli Amson², Aldo Benites-Palomino³, Giorgio Carnevale⁴, Giovanni Coletti⁵, Alberto Collareta¹, Claudio Di Celma⁶, Olivier Lambert⁷, Elisa Malinverno⁵, Luca Mariani⁸, Giuseppe Marramà⁴, Stefano Mazzoli⁶, Federica Mulè¹, Francesco Nobile¹, Luca Pellegrino⁴, Pietro Paolo Pierantoni⁶, Mario Urbina⁹, Rafael Varas-Malca⁹, Giovanni Bianucci¹

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⁹ Departamento de Paleontología de Vertebrados, Museo de Historia Natural-UNMSM, Peru

The East Pisco Basin (EPB) preserves an exceptional fossil record documenting major paleoclimatic and paleoceanographic changes associated with the evolution of the Humboldt Current System. Paleontological evidence from the Chilcatay Formation (Early Miocene) and the lower Pisco Formation sequence (P0; Middle Miocene) reveals warm-water coastal environments characterized by thermophilic faunas. Invertebrate assemblages include thermophilic taxa such as the gastropods *Muracypraea ormenoi* and *Ficus distans*, specimens of the family Architectonicidae, and a rhizangiid coral resembling the (sub)tropical genus *Culicia*. Vertebrate assemblages comprise warm-water elasmobranchs such as the sawfish *Anoxypristis* sp. and the weasel shark *Hemipristis serra*, together with other mesopredators indicating tropical to subtropical coastal conditions. In contrast, the upper Pisco Formation (P1–P2; Late Miocene) record the establishment of highly productive upwelling conditions. These younger assemblages are characterized by abundant mysticete whales, odontocetes, pinnipeds, seabirds, sharks, rays, and schooling fishes such as sardines, forming a tropically complex ecosystem supported by high primary productivity. The increasing abundance of filter-feeding whales and pelagic vertebrates, together with sedimentological

evidence such as diatom-rich deposits characterized by upwelling species, indicates intensified coastal upwelling and cooler oceanographic conditions. The fossil record of the EPB captures the past evolution of the modern Humboldt Current Ecosystem, highlighting the role of oceanographic cooling and upwelling intensification in shaping marine biodiversity along the southeastern Pacific margin.

Poster 2: GARRIDO

The Warm, the Cool, and the Deep: Contrasting deep-water oxygenation during MIS 9 and MIS 13 in the Southeast Pacific

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Ocean deoxygenation threatens marine ecosystems by affecting biodiversity, nutrient cycling, and the global carbon cycle. The Southeast Pacific (SEP) hosts one of the world's largest oxygen-deficient zones (ODZs), yet the response of its deep waters to past climate change remains poorly understood. Studying deep-water oxygenation during past climate variability provides valuable analogues for future deep-ocean deoxygenation. Here, we reconstruct bottom-water dissolved oxygen (BWDO) at ODP Site 1237 (3,200 m depth) during two contrasting Pleistocene interglacials: Marine Isotope Stage (MIS) 9, the warm interglacial, and MIS 13, the cool interglacial. BWDO was reconstructed using a multiproxy approach based on benthic foraminiferal shell porosity and carbon isotope gradients. Our results show pronounced deoxygenation during the warm MIS 9 (~330–300 ka), with BWDO declining below 100 $\mu\text{mol kg}^{-1}$. In contrast, the cooler MIS 13 (~500–470 ka) reached BWDO values exceeding 200 $\mu\text{mol kg}^{-1}$ before declining towards the onset of MIS 12. Our findings suggest that warm and cool interglacials caused different deep-water oxygenation states in the Southeast Pacific, providing long-term analogues for understanding future deep-ocean deoxygenation amid ongoing global warming.

Poster 3: Paleo3 Journal – Special Issue

From Land to Ocean: South American climate variability from the Miocene to Holocene (Editorial preface)

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South America features some of the most dynamic and climatically diverse environments on Earth; however, its palaeoenvironmental records remain sparse and unevenly distributed. Such data scarcity limits our understanding of South America's critical role in driving global climate dynamics, ocean-atmosphere interactions, and regional biodiversity patterns.

This Virtual Special Issue (VSI) entitled *From Land to Oceans: South American climate variability from the Miocene to Holocene* brings together 14 papers that investigate atmospheric, hydrological, and oceanographic variability across a broad range of temporal scales. The contributions are organized into three categories. The first focuses on Neogene to Early Quaternary evolution, addressing long-term changes in ocean circulation and major climate systems. The second examines Late Quaternary and deglacial variability, emphasizing glacial-interglacial transitions and hydroclimatic shifts. The third explores Late Holocene climate variability through high-resolution reconstructions spanning the last few millennia. These studies provide new insights into the mechanisms driving climate variability across South America and highlight the value of integrating marine and continental archives to better understand past environmental change.

We emphasize that this VSI reflects a diverse and inclusive scientific effort, bringing together contributions from early-career researchers, female scientists, and a broad range of South American institutions.

Poster 4: ROCO-VILLABLANCA

Scaphopoda of the Miocene of Chile

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Scaphopods (Class Scaphopoda, Phylum Mollusca), commonly known as tusk shells, are exclusively marine mollusks that live buried within soft substrates. This class comprises two orders: Dentaliida, characterized by elongated shells with sculpture and ornamentation, and Gadilida, which includes smaller and smoother shells.

A total of 2,103 scaphopod specimens from Lower Miocene (Burdigalian) marine deposits of the Navidad Formation, Ranquil Formation, Lacui Formation, and Ipún beds were analyzed in order to identify species and evaluate paleobiogeographic patterns along a latitudinal transect between 33°S and 45°S. Species diversity was assessed using alpha, beta, and gamma diversity metrics, as well as rarefaction analyses.

Eleven scaphopod species were identified from the Lower Miocene of Chile, including the first gadilid records reported for the Chilean Miocene. The results show no significant latitudinal species turnover among the studied localities, suggesting the existence of a single paleobiogeographic province during this time interval based on the scaphopod fauna. These findings provide new insights into the early Miocene marine biodiversity and paleobiogeographic structure of the southeastern Pacific during the early stages of the Humboldt Current System.

Poster 5: GUERRERO

Paleobiogeographic patterns of Lower Miocene (17-18 Ma) marine faunas from south-central Chile.

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In the southeast Pacific, the Peruvian warm-temperate (6°S – 42°S) and Magellanic cool-temperate biogeographical provinces (42°S – 56°S) are relatively well recognized in modern marine faunas. During the Lower Miocene (17 – 18 Ma), the occurrence of tropical taxa at mid-latitudes, associated to warmer global conditions and lower latitudinal thermal gradient could indicate a different biogeographical structure along the coast of Chile.

Here, we present preliminary results on the latitudinal paleobiogeographic patterns of Lower Miocene bony fishes, bivalves and gastropods from the Navidad Formation (~33°S), Ranquil Formation (~36°S), Lacui Formation (~42°S) and Ipún beds (~44°S) in south-central Chile. We use a database of 415 taxa and 36.411 specimens to calculate dissimilarity matrices and assess faunal differences between localities and geological units. Matrices were subjected to hierarchical clustering and ordination analysis to test the hypothesis about a homogeneous fauna across the latitudinal gradient. Dissimilarity values are relatively low between localities of the four geological units and show no major latitudinal break. In both analysis, localities of lower and higher latitudes are mixed, suggesting faunal linkages across the whole latitudinal gradient. These results could reflect a weak paleobiogeographical significance of the Humboldt Current System, derived from a different paleoclimatic setting in the southeast Pacific.

Poster 6: SUSAYA-VICENTE

Environmental Gradients Shape Functional Diversity in Living Benthic Foraminifera

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The Humboldt Current System in the southeastern Pacific is characterized by strong environmental gradients associated with oxygen minimum zones, high productivity, and complex water mass structure, all of which strongly influence benthic ecosystem functioning. In this study, living benthic foraminiferal communities sampled between 1°S and 50°S are used as a biological model to evaluate how environmental conditions shape functional diversity across this margin. Functional traits describing test morphology, feeding strategy, microhabitat preference, and oxygen tolerance were assigned to species. Functional diversity indices and community-weighted means were calculated to characterize community functional structure. These metrics were related to bottom-water properties including depth, temperature, salinity, oxygen and nutrients, and sediment geochemistry including organic carbon, nitrogen, biogenic silica, carbonate content using generalized additive models. Results show clear spatial structuring of functional diversity across the region. Oxygen availability emerges as the main driver of composition, with low-oxygen environments favoring opportunistic strategies and reduced functional diversity, whereas well-oxygenated settings support higher functional heterogeneity and broader functional traits. Sediment geochemistry and water mass properties further modulate these patterns. These results highlight environmental filtering and demonstrate value of functional approaches using living benthic foraminifera to understand ecosystem responses in oxygen-deficient eastern boundary upwelling systems.

Poster 8: DeVRIES

The "Oligocene Gap" in the East Pisco Basin (Peru): An Opportunity to Predict the Paleogene-to-Neogene Evolution of the Humboldt Current System

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The Neogene marine microfossil record of the East Pisco Basin (EPB) in southern Peru is well studied and has become the basis for recognizing step-wise intensification of the Humboldt Current System (HCS) and/or attendant coastal upwelling since the Early Miocene. An Early Oligocene to earliest Miocene (32–22 Ma) gap in our knowledge of the paleoenvironmental record of the EPB has hindered our understanding of the connection between conditions of the Neogene HCS and antecedent Paleogene conditions of initially temperate waters followed by cooling and increasing productivity across the Eocene-Oligocene Transition. Only a single sample from the EPB – fine-grained and diatomaceous (~27–26 Ma) – has been found that could give insight regarding the behavior of the HCS during the Oligocene to earliest Miocene. Fossils from "Oligocene gap" strata might be expected to document shifts from low latitude to higher latitude affinities of marine mollusks and replacement of archeocete cetaceans and stem groups of mysticetes with odontocetes and crown groups of mysticetes.

An unusual opportunity exists to posit paleontological scenarios regarding the Paleogene-to-Neogene transition of the HCS prior to undertaking further fieldwork to find the missing Oligocene sequence and test those scenarios.

Poster 9: PRASANNAKUMAR

The Evolution of Humbolt current system in AWI-ESM

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The Humboldt Current System (HCS) is considered to play a fundamental role in shaping the hydroclimate of the Atacama Desert, one of the driest regions on Earth. As part of the CRC 1211 project *Earth – Evolution at the Dry Limits*, we investigate the land–ocean coupling mechanisms that influence hydroclimatic variability in the Atacama region.

In this study, we explore the evolution of the Humboldt Current System using the Earth System Model AWI-ESM. Simulations are conducted for four distinct climate states: the Miocene, the Last Glacial Maximum, the Last Interglacial, and the Mid-Holocene. AWI-ESM couples the atmospheric model ECHAM with the ocean–sea ice model FESOM and includes the Water Isotope (WISO) module, which simulates the absolute concentrations of water isotopes in the atmosphere, ocean, and sea ice. The unstructured mesh of FESOM enables enhanced spatial resolution in dynamically active coastal regions, providing a powerful framework for investigating the evolution of the Humboldt Current System under different climatic boundary conditions. We analyze changes in the HCS across these climate states and evaluate the model performance against proxy reconstructions of sea-surface temperature.

Poster 10: SALDIVIA

Bathymetric changes in the Seno Hoppner Formation, Southern Chile, across the Pliocene–Pleistocene boundary based on benthic foraminifera

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In this study, we analyzed fossil foraminifera assemblages from the Seno Hoppner Formation, spanning the Pliocene–Pleistocene boundary on the Taitao Peninsula (46.12°S), Chile. A total of 54 taxa were identified, comprising 46 benthic and 8 planktonic. The objective was to reconstruct the bathymetry of the depositional environment based on the occurrence of deep-water indicator species rather than on the relative abundances of shallower taxa. This approach considers the tectonic setting of the area, characterized by frequent sediment erosion and redeposition, which facilitate the mixing of shallow and deep–water taxa within the same deposit. The results indicate that relative abundances do not adequately reflect the bathymetric conditions of the basin. Nevertheless, the assemblages suggest a shelf environment (0–200 m water depth), with a transition from outer to inner-shelf from west to east, which is consistent with rapid basin uplift. Although it's not possible to establish a robust biostratigraphic correlation among the studied sites, the assemblages are consistent with the Pliocene–Pleistocene boundary interval, as indicated by the occurrence of *Notorotalia mangaoparia* and *N. zelandica*. Additionally, the composition of the planktonic assemblage and the high abundance of taxa belonging to the family Cassidulinidae point to cold-water paleoenvironmental conditions.

8. Debate Sessions

Six 60-minute debate sessions are included in the program, providing an opportunity for guided discussions on specific topics that we hope will help shape the future of HCS research. At the same time, these discussions aim to build a strong and active scientific network that can collectively work towards common goals and shared interests. The network should foster collaboration and make the best possible use of the complementary resources, expertise, and capabilities available among its members.

As a reminder, we expect these sessions to be **spaces of respect, inclusion, and constructive scientific dialogue**. Participants should feel comfortable sharing ideas and perspectives, while respecting the ideas, contributions, and intellectual ownership of others. Discriminatory, offensive, or dismissive behavior will not be tolerated.

Day 1 (Wednesday, Sept 23)

Debate 1: Ocean–Climate Dynamics (2:40 PM) and Debate 2: Record Gaps (4:30 PM)

Objective: To discuss the principal mechanisms controlling the long-term evolution of the Humboldt Current System and identify the most pressing scientific questions that require collaborative investigation to fill data gaps (record gaps).

Key Questions

- How can we define the HCS in the geological record?
- What were the principal drivers of long-term changes in circulation and upwelling intensity? What are the dominant controls on HCS variability from the Miocene to the present?
- Which geological and paleoceanographic records are needed to provide evidence or to explain these changes?
- Where do significant spatial and temporal gaps remain in the available proxy records?
- Which hypotheses regarding the HCS remain unresolved and/or untested?
- Which datasets are required to test specific hypotheses?

Expected Outcomes

- A synthesized overview of the current understanding of HCS oceanographic evolution.
- Identification of priority proxy and geographical data gaps.
- A prioritized list of scientific questions.
- Establishment of a common scientific framework for discussions during the following workshop sessions.
- Identification of opportunities for collaborative studies.

Day 2 (Thursday, Sept 24)

Debate 3: Ecological Dynamics (2:00 PM)

Objective: To evaluate current understanding of ecosystem responses and resilience to long-term climate and environmental changes and to identify opportunities for integrating biological, paleontological, geological, and geochemical evidence.

Key Questions

- How have HCS ecosystems responded to changes in ocean circulation and climate through time?
- What are the links between physical oceanographic processes and the evolution of ecosystems and biodiversity?
- Which ecological proxies best capture past environmental variability?
- How can different proxy records be integrated to reconstruct ecosystem dynamics?
- What new developments regarding proxies, data mining and/or machine learning are needed?
- How well can marine and terrestrial records be integrated? Can terrestrial ecosystem be used to understand marine processes and vice versa?

Expected Outcomes

- A synthesized overview of the current understanding of the HCS coastal-marine ecological dynamics.
- Improved integration between oceanographic and ecological perspectives.
- Development of interdisciplinary priorities for future ecological and paleo(biological) studies within the HCS.
- Identification of complementary approaches and opportunities for collaborative research.

Debate 4: Funding efforts (4:20 PM)

Objective: To discuss opportunities and strategies for developing collaborative funding efforts that can support research on the HCS, with particular emphasis on interdisciplinary projects, shared infrastructure and datasets, and the scientific priorities identified during the workshop. **This is the first of two debate sessions with complementary objectives.**

Key Questions

- Which scientific questions or research themes are best suited to collaborative, interdisciplinary funding proposals?
- What are the main funding needs, common interests, and priorities for advancing research on the HCS?

- What national, international, and regional funding opportunities could support joint HCS research?
- Could the HCS community develop common project ideas, funding priorities, or proposal frameworks that can be adapted to different funding calls? Is this of interest within the HCS community?

Expected Outcomes

- Identification of priority research areas suitable for collaborative funding efforts.
- Overview of relevant funding opportunities and mechanisms for international collaboration.
- Identification of complementary resources, expertise, infrastructure, and datasets available within the network.
- Identification of potential project ideas, partnerships, and consortium structures.

Day 3 (Friday, Sept 25)

Debate 5: Proxy-Model Integration

Objective: To identify the main challenges and opportunities in combining proxy records with numerical models to reconstruct the evolution of the HCS and surrounding terrestrial landscapes.

Key Questions

- Where do proxy reconstructions and model simulations agree or diverge? Where are the largest uncertainties?
- What are the major uncertainties limiting proxy–model comparisons?
- Which proxy records are most suitable for model evaluation? Which proxy datasets are most useful for model validation?
- What coordinated efforts could improve future comparisons?
- How can the modelling efforts be integrated with international initiatives such as PlioMIP or DeepMIP?
- How can new collaborations improve future reconstructions and predictions?

Expected Outcomes

- Identification of priority datasets for model evaluation.
- Identification of priority proxy–model integration challenges.
- Recommendations for future proxy–model collaborations and coordinated data generation
- Design of a broad yet realistic roadmap for future interdisciplinary research.

Potential Outcomes

- Overview of relevant funding opportunities and mechanisms for international collaboration.
- Identification of complementary resources, expertise, infrastructure, potential project ideas, partnerships, and consortium structures.
- Strengthening of the network's capacity to coordinate and support joint funding efforts.

- **From Cologne Central Station (*Köln Hbf*):** take a train to Cologne South station (*Köln-Süd*) (about 7 minutes), then walk a short distance to Zülpicher Straße 49b. You can also take tram line 18 toward "*Klettenberg/Bonn*" and get off at station "*Eifelwall/Stadtarchiv*" or use Tram line 9 toward "*Sülz*" and get off at station "*Dasselstraße/Bf Süd*".

- From Düsseldorf Airport (DUS): take any train toward Cologne Central Station (*Köln Hbf*), then continue by regional train to Cologne South station (*Köln-Süd*) or by tram as above; total travel time is around 60 minutes depending on the connection.

Alternatively, you can connect through a **guest WiFi pass** available at the registration desk.

Social Dinner (Day 1 – Wednesday, 23 September): The dinner will take place on-site in the institute's garden. Vegetarian and non-vegetarian food and drinks will be provided, including local beer.

- Ambulance & fire department: **112**

10. Acknowledgements

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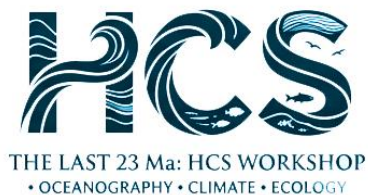
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